

FracVAL: A Simplified Tunable Code to Generate Fractal Structures by CC Aggregation of Polydisperse Primary Particles

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Abstract

Fractal aggregates formed by spherical polydisperse nano-sized primary particles are important in various fields of science and engineering. Primary particle polydispersity has been found to significantly influence the kinetics of aggregation, sintering and the properties of fractal aggregates. Tunable algorithms are a powerful numerical tool to generate fractal aggregates that possess the prescribed fractal dimension D_f and prefactor k_f for each individual aggregate and are ideal to isolate the effect of D_f and k_f on the properties of fractal aggregates. In this study, the tunable algorithm of cluster-cluster aggregation developed by Filippov et al. 2000 for generating fractal aggregates formed by monodisperse spherical primary particle is extended to polydisperse primary particles. This new algorithm, termed FracVal, is developed by using the recursive mathematical solution to the intersection points of two spheres. The application of this concept allowed us to find the exact mass center position of each monomer and similarly the cluster to be aggregated to an existing aggregate. The algorithm is able to preserve the prescribed fractal dimension and prefactor for each aggregate, regardless of its size, with negligible error for lognormally distributed primary particles with the geometric standard deviation $\sigma_{p,geo}$ being as large as 3. In contrast, for polydisperse primary particles the direct use of Filippov et al. 2000 method, as is done by Skorupski et al. 2014, does not ensure the preservation of D_f and k_f for individual aggregates and it is necessary to generate a large number of aggregates to achieve the prescribed D_f and k_f on an ensemble

basis. The performance of FracVAL is evaluated for aggregates consisting of 500 monomers and for fractal dimension variation over the entire range of $D_f > 1.0$ and < 3.0 and the prefactor within the $k_f = 0.1-2.7$ range. All aggregates are generated on average in less than 2.4 minutes on a common laptop, illustrating the efficiency of the proposed algorithm.

Keywords: Tunable Algorithm; Fractal Aggregates; Polydisperse Primary Particles; Aggregation

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1. What is FracVAL?

FracVAL consists of a numerical environment designed for the simulation of fractal-like aggregates with specific morphology. In the current version (**FracVAL_v1**) aggregates consist of point-touching, spherical polydisperse primary particles.

FracVAL: A number of sub-clusters are generated by using a particle-cluster aggregation subroutine, then a hierarchical cluster-cluster aggregation takes place.

1.1. Aggregates morphology

For an aggregate consisting of point-touch polydisperse PPs, its total mass (m_a) follows fractal scalling law (Eggersdorfer and Pratsinis 2012¹),

$$\frac{m_a}{\overline{m}_p} = k_f \left(\frac{R_g}{r_{p,geo}} \right)^{D_f}, \quad (1)$$

where, in this study $\overline{m}_p$ is the geometric average PP mass, consequently $m_a/\overline{m}_p = N$ the number of PP. The proportionality constant k_f is called fractal prefactor, $r_{p,geo}$ is the geometric average PP size and R_g is the radius of gyration of the aggregate calculated as proposed by Dastanpour et al. 2016², which is still valid in the limit $N \sim 1$,

$$R_g^2 = \frac{1}{m_a} \sum_{i=1}^N m_{p,i} \left[(R_i - R_c)^2 + \frac{3}{5} r_{p,i}^2 \right], \quad (2)$$

1.2. Primary particle size distribution

Eq. (3) presents the probability density function of PP radii where $f[r_p]dr_p$ is the probability of finding a particle with a radius between r_p and $r_p + dr_p$,

¹Eggersdorfer, M. L., & Pratsinis, S. E. (2012). The structure of agglomerates consisting of polydisperse particles. *Aerosol science and technology*, 46(3), 347-353.

²Dastanpour, R., & Rogak, S. N. (2016). The effect of primary particle polydispersity on the morphology and mobility diameter of the fractal agglomerates in different flow regimes. *Journal of Aerosol Science*, 94, 22-32.

$$f[r_p] = \frac{N}{\ln[\sigma_{geo}]\sqrt{2\pi}r_p} \exp \left[-\frac{1}{2} \left(\frac{\ln[r_p] - \ln[r_{p,geo}]}{\ln[\sigma_{geo}]} \right)^2 \right], \quad (3)$$

additionally, $r_{p,geo}$ and σ_{geo} correspond to the geometric mean and standard deviation of PP radii.

1.3. Main equation

If two clusters of mass m_1 and m_2 , radius of gyration R_{g1} and R_{g2} have the same D_f and k_f and we want to aggregate them preserving these fractal parameters then, they should be placed at a distance Γ , which value is given by the following equation,

$$m^2 R_g^2 = m (m_1 R_{g1}^2 + m_2 R_{g2}^2) + \Gamma^2 m_1 m_2, \quad (4)$$

where $m = m_1 + m_2$ and R_g are the mass and radius of gyration of the resulting aggregate. A detailed derivation of Eq. (4) is provided in the Appendix of the paper we published this algorithm.

2. FracVAL

2.1. Architecture of the algorithm

The algorithm consists in 7 modules as shown in Figure ???. These modules are described as follows,

- **FracVAL_CCA:** This is the main module of FracVAL, its goal is to do an iterative process until generating the required number of aggregates defined by the user.
- **Ctes:** This module contain all the parameters used by the algorithm.
- **CCA:** This is the core module of FracVAL. Its aim is the generation of one aggregate at the time by following an iterative process as described in the manuscript.
- **PCA_Subclusters_module:** This is the main module of the particle-cluster subroutine explained before, its goal is to do an iterative process until generating the required number of sub-clusters defined by the user.

- PCA_cca: This is the core module of the particle-cluster subroutine. Its aim is the generation of one sub-cluster at the time by following an iterative process as described in the manuscript.
- Random_PP: Generate a specific quantity of numbers by using the module called “random”.
- random: Contain the a pseudo-random number generator where the user can select a specific probability density function.
- RAND_SAMPLE: A module that contain a function used for the random sampling of one value from a vector or matrix.

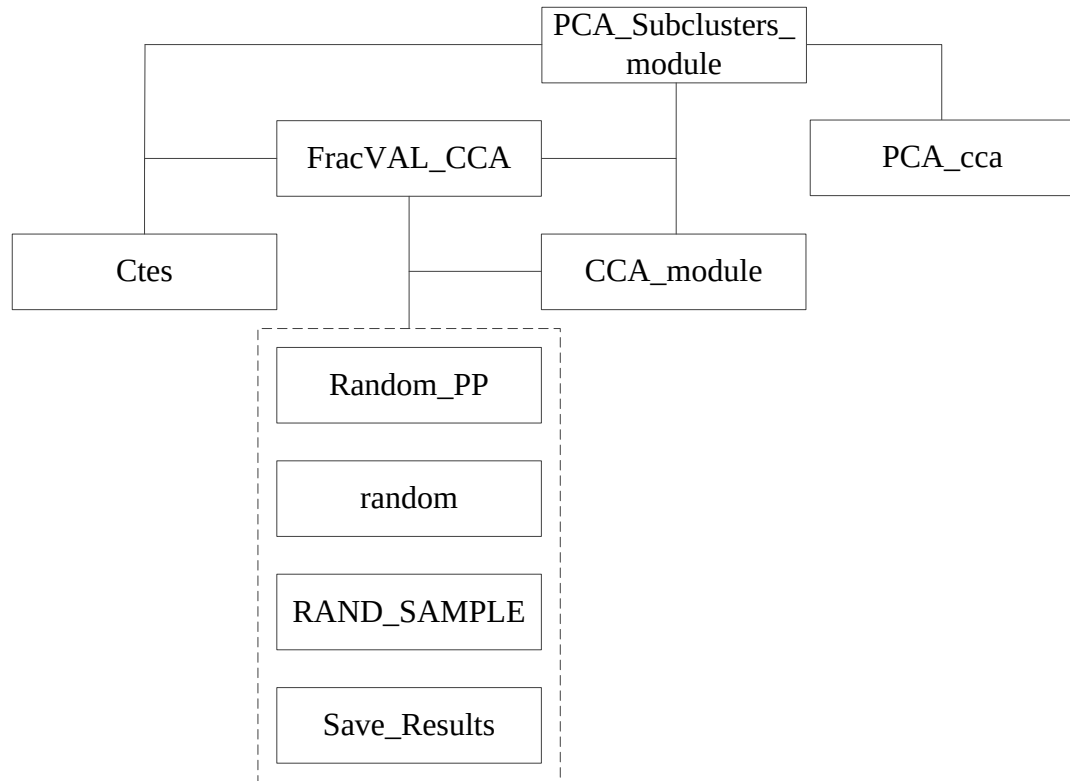


Figure 1: Architecture of FracVAL.

2.2. Module Ctes

This is the most important module for the user, because it is used for the definition of all the parameters and main variables used by the algorithm.

```

1 module Ctes
2   implicit none
3   integer, parameter:: N=500                                !Number of PP
4   real, parameter:: Df=1.79                                  !Fractal dimension
5   real, parameter:: kf=1.40                                  !Fractal prefactor
6   real, parameter:: rp_g=15.0                                !Geometric mean PP
7   real, parameter:: rp_gstd=1.00                             !Geometric PP standard deviation
8   integer, parameter:: Quantity_aggregates = 50             !Quantity of aggregates to be generated
9   integer, parameter:: Ext_case = 0                          !Activate extreme cases
10  real, parameter:: Nsubcl_perc = 0.1                        !controls the subclusters size (keep always in 10%,
                                                                !only increase when PC is not able to work)
11                                                                !Tolerance to overlapping
12  real, parameter:: tol_ov=10.**(-6.)
13  real, parameter :: pi=4.0*atan(1.0)
14  real R(N)                                                    !Primary particles radii and mass
15  real X(N), Y(N), Z(N)                                         !Coordinates of primary particles
16  integer:: iter, i, j, k
17 end module Ctes

```

Figure 2: Parameters and variables of the module Ctes. Screenshot taken on the interface of Eclipse (<http://www.eclipse.org>).

The parameters related with aggregates morphology are described as follows,

- N: The number of monomers in an aggregate. Corresponding to the ratio $m_a/\overline{m_p} = N$ in the LHS of Eq. (1).
- Df: The fractal dimension. Corresponding to the exponent D_f in RHS of Eq. (1).
- kf: The fractal prefactor. Corresponding to the prefactor k_f in RHS of Eq. (1).

Regarding primary particle polydispersity,

- rp_g: Geometric primary particle mean. Corresponding to $r_{p,geo}$ in Eq. (3).
- rp_gstd: Geometric primary particle standard deviation. Corresponding to σ_{geo} in Eq. (3).

Other parameters,

- `Ext_case`: Binary parameters used to activate or deactivate “extreme cases”. See sub-step 8-b in the manuscript.
- `Nsubcl_perc`: Number of monomers of each sub-cluster. It is used by the module “PCA_cca”.
- `Quantity_aggregates`: Defines the number of aggregates to be generated by running once the module “FracVAL_PCA”.
- `tol_ov`: A tolerance value used by the module “PCA” to ensure that primary particles will be in one point-contact.

Main variables,

- `X,Y,Z`: Three vectors of length `N`, corresponding to the Cartesian coordinates of each monomer.
- `R,M`: Two vectors of length `N`, corresponding to the radius and mass of each monomer.

2.3. Module CCA

The FracVAL (fractal aggregate generation algorithm developed in Valparaíso) algorithm is programmed in a hierarchical manner, i.e., only aggregation between sub-clusters with approximately the same number of primary particles is allowed. The algorithm consists of four main steps corresponding to the flow chart shown in Fig. 3 and it is described as follows. More details are given in the manuscript.

3. Installation and execution

3.1. Installation

FracVAL was programmed in the language FORTRAN 90, i.e. a Fortran compiler is required. We recommend using gfortran³.

3.2. Execution

For both Windows and Linux the programs can be executed by using an IDE such as Eclipse or Netbeans, or also by using the respective terminal. Note that always gfortran compiler should be installed in the computer (see section 3.1). Here we will describe the execution by using the terminals.

³Free available from: <https://gcc.gnu.org/wiki/GFortranBinaries>

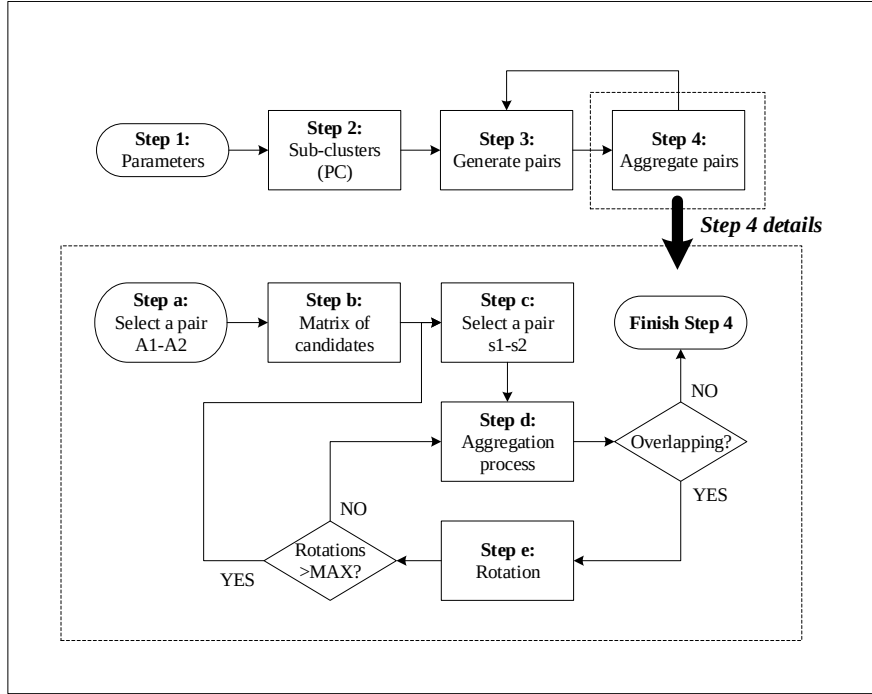


Figure 3: Flow chart of FracVAL cluster cluster aggregation algorithm.

3.2.1. Execution in Windows

1. Open the cmd Terminal
2. Search for the containing folder:
C:\ ... \ FracVAL_v1\ PROGRAM_source_codes\ Windows
3. Write the following commands for compilation: gfortran Frac_VAL_CCA.f90
Ctes.f90 CCA_module.f90 PCA.cca.f90 PCA.Subclusters_module.f90 RAND_SAMPLE.f90
random.f90 Random_PP.f90 Save_results_CC.f90 -o FRACVAL_CCA_executable.exe
IMPORTANT: If a “.mod” file is not generated, then TRY AGAIN.
4. Once all the required “.mod” have been generated. Write the following
commands for execution: FRACVAL_CCA_executable.exe

3.2.2. Execution in Linux

1. Open the Ubuntu’s Terminal
2. Search for the containing folder:
C:/.../FracVAL_v1/PROGRAM_source_codes/Linux
3. Write the following commands for compilation: gfortran Frac_VAL_CCA.f90
Ctes.f90 CCA_module.f90 PCA.cca.f90 PCA.Subclusters_module.f90 RAND_SAMPLE.f90

random.f90 Random_PP.f90 Save_results_CC.f90 -o FRACVAL_CCA_executable

4. Write the following commands for execution: ./FRACVAL_CCA_executable

4. Understanding the output files

The results are created as .DAT files. Each column correspond to each monomer. The first 3 columns correspond to the X,Y,Z coordinates respectively. The column 4 corresponds to the corresponding radius of each monomer. In Figure 4 an example of output file is provided, as can be seen this is an aggregate consisting of 24 monodisperse monomers of radius 15.

1	4.27577400	26.4779644	56.5807190	15.0000000
2	-1.29627037	52.3628769	70.6848145	15.0000000
3	-23.6242008	16.3502331	52.2194633	15.0000000
4	22.7405472	4.11893272	48.8914452	15.0000000
5	7.22603226	70.9525681	48.7350159	15.0000000
6	-40.6996346	12.1230488	76.5209045	15.0000000
7	-13.3242722	77.4326859	27.8617516	15.0000000
8	13.7878609	-13.4193478	26.2583103	15.0000000
9	27.3345451	86.7080307	64.4643860	15.0000000
10	-39.7803764	90.6505661	32.8978577	15.0000000
11	-41.4293594	-17.9330502	134.889542	15.0000000
12	-41.9815369	-44.8348160	121.623413	15.0000000
13	-54.5709801	-1.04290771	113.865250	15.0000000
14	-21.2797165	-6.62741280	154.025208	15.0000000
15	-65.1069717	-17.4526653	91.0680389	15.0000000
16	-15.4127312	20.7186565	143.172958	15.0000000
17	-69.8745575	11.1676998	83.4427643	15.0000000
18	-83.1346054	-40.9394684	86.2332611	15.0000000
19	-24.6664200	-28.2346554	174.559509	15.0000000
20	-13.0421562	5.60710716	180.149124	15.0000000
21	-70.4661713	-39.7571716	-3.24190140	15.0000000
22	-42.3468590	-31.3391132	2.95817566	15.0000000
23	-65.8668365	-67.5952682	-13.4343643	15.0000000
24	-20.1933861	-50.5664444	9.24573517	15.0000000

Figure 4: Example of output file of FracVAL (Picture from Notepad++).

5. Visualizing the results

In the folder “PROGRAM_visualizing_results” you will find a Matlab subroutine called “Visualizing_results”. By executing that file then you will

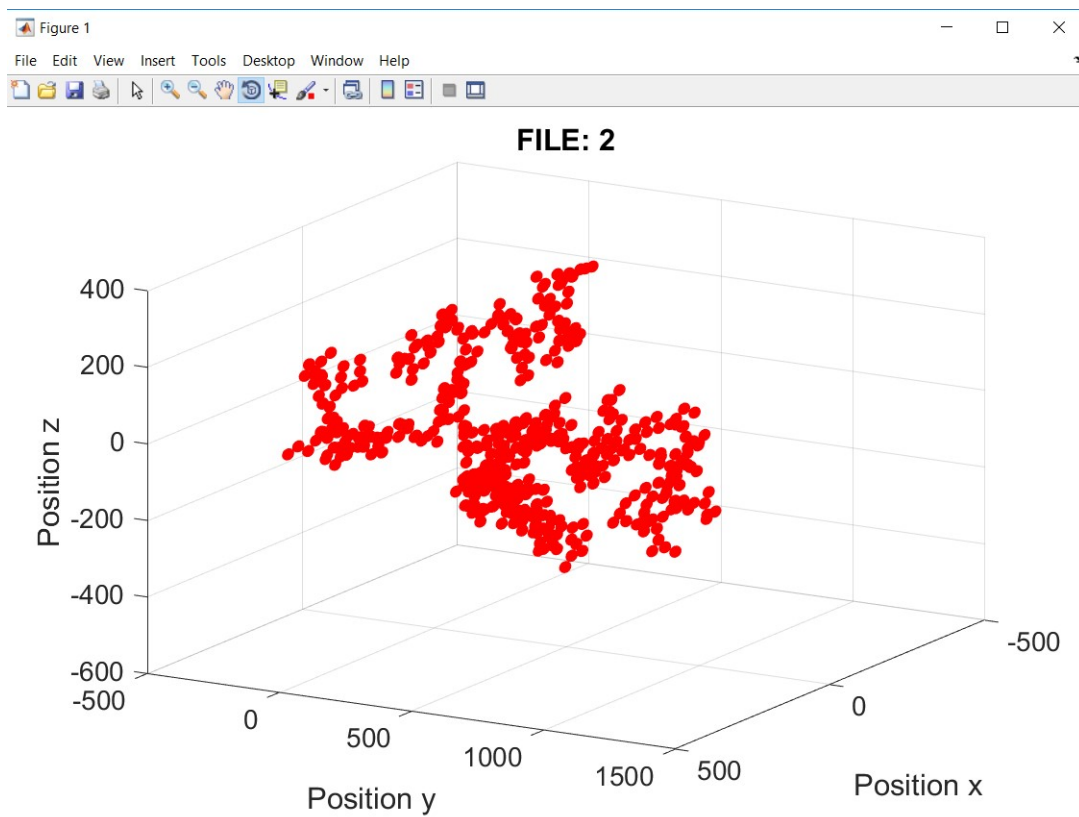


Figure 5: Example of aggregate visualized by using Matlab.

be able to visualize and analyze each aggregate. You should see a plot of the aggregate as shown in the following screen shot,

6. Solving problems

There are some cases difficult to model by FracVAL.

- PC is not able to generate sub-clusters: Try the following option:
 - Increase the value of the variable `Nsubcl_perc`. It is recommended to be as low as possible.
 - Activate extreme cases by switching `Ext_case = 1` (for example, for extreme fractal parameters.)

- CC is not able to generate clusters: Check the combination of fractal parameters. Sometimes it only needs one additional compilation.

7. Examples of aggregates

$$N_p=1024, \sigma_{p,geo}=1.00$$

$$D_f=1.79, k_f=1.40$$

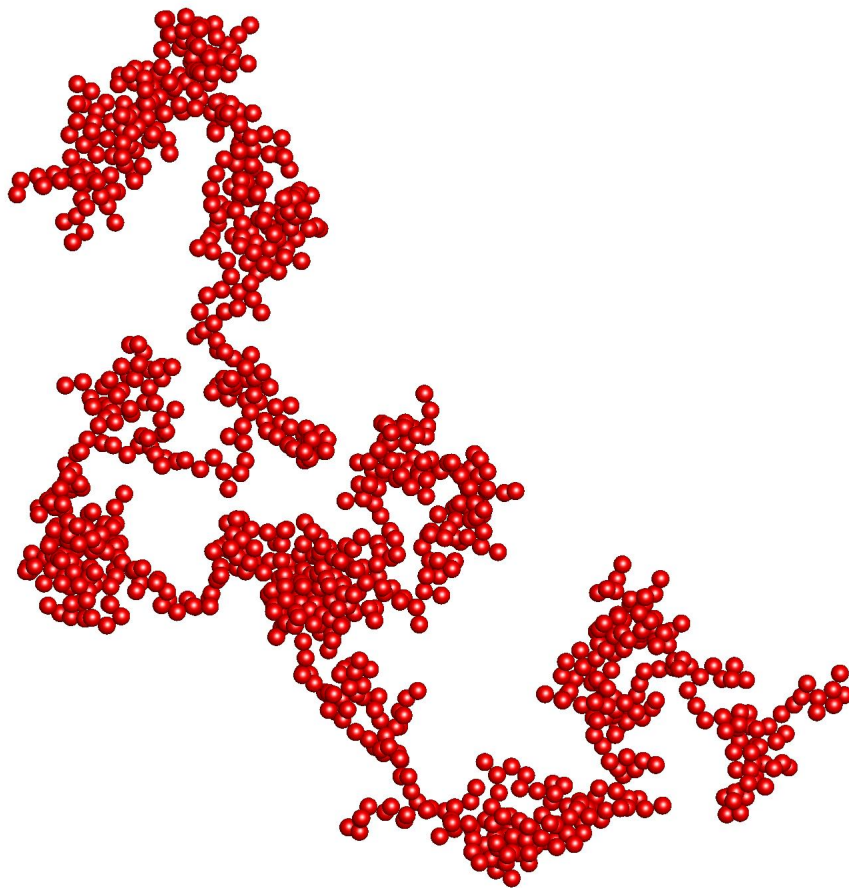


Figure 6: Example of aggregate consisting of $N=1024$ monomers and $\sigma_{p,geo} = 1.00$.

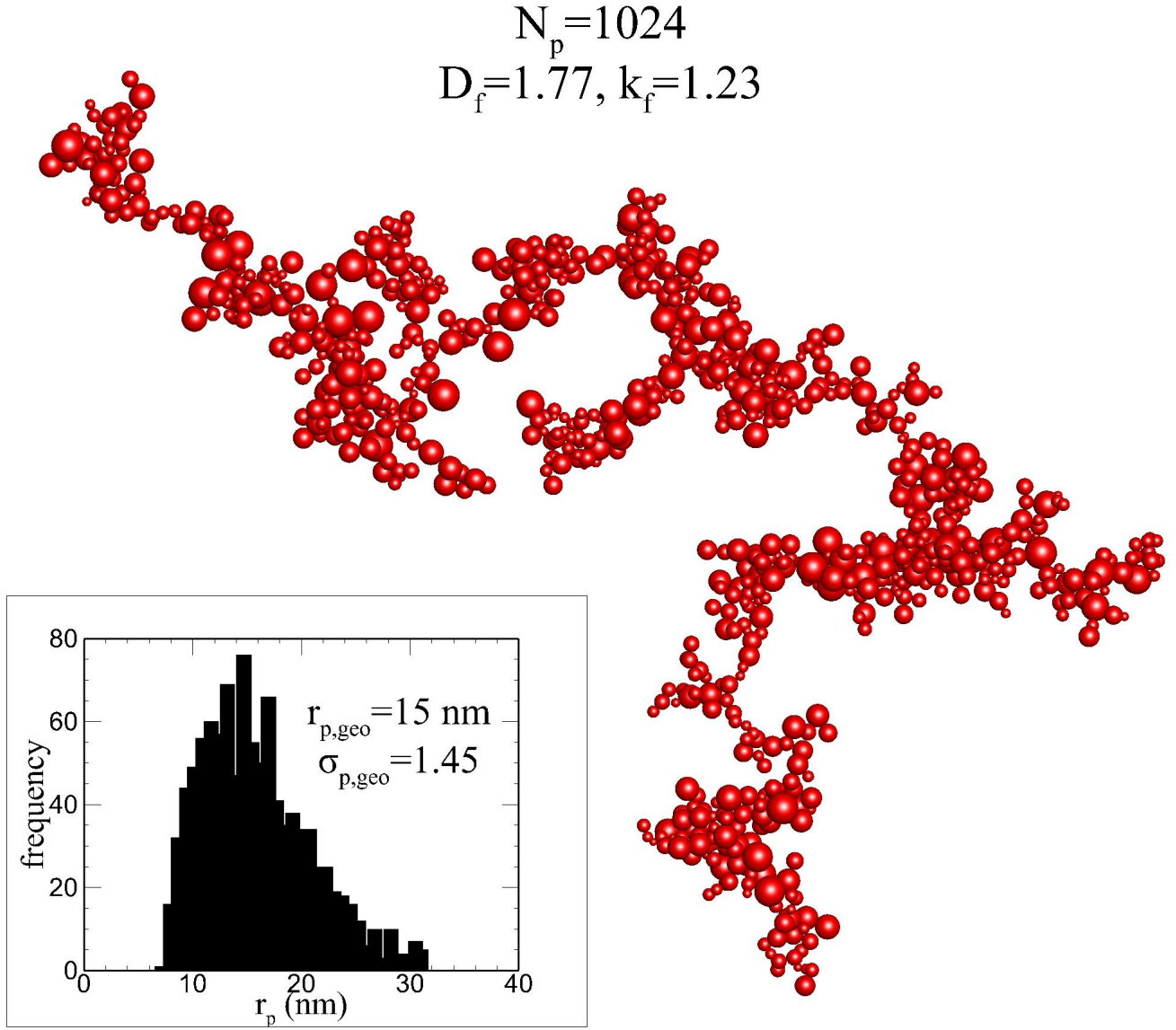


Figure 7: Example of aggregate consisting of $N=1024$ monomers and $\sigma_{p,geo} = 1.45$.

$$N_p=1024$$

$$D_f=1.68, k_f=0.98$$

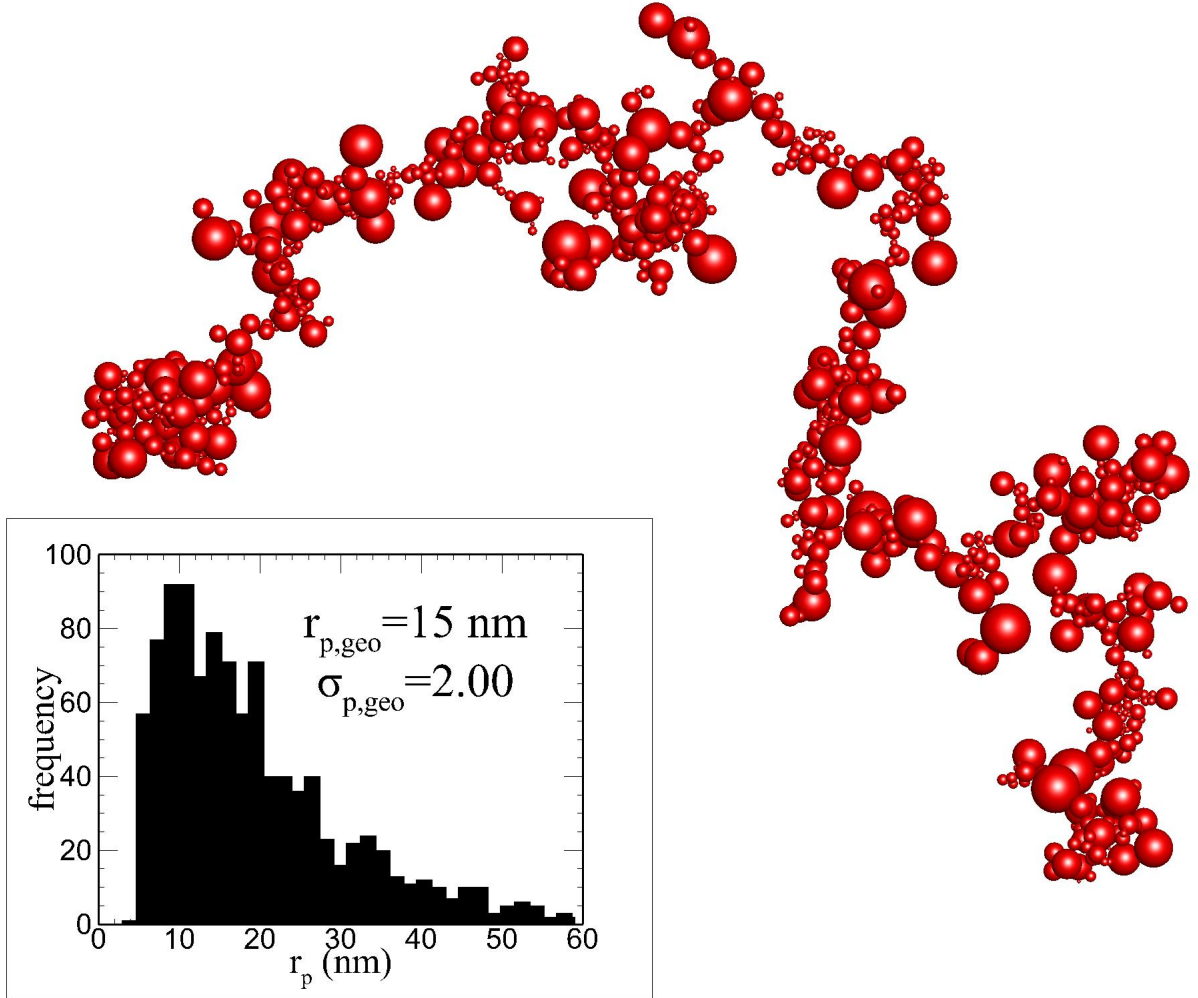


Figure 8: Example of aggregate consisting of $N=1024$ monomers and $\sigma_{p,geo} = 2.00$.

$$N_p=1024$$

$$D_f=1.57, k_f=0.90$$

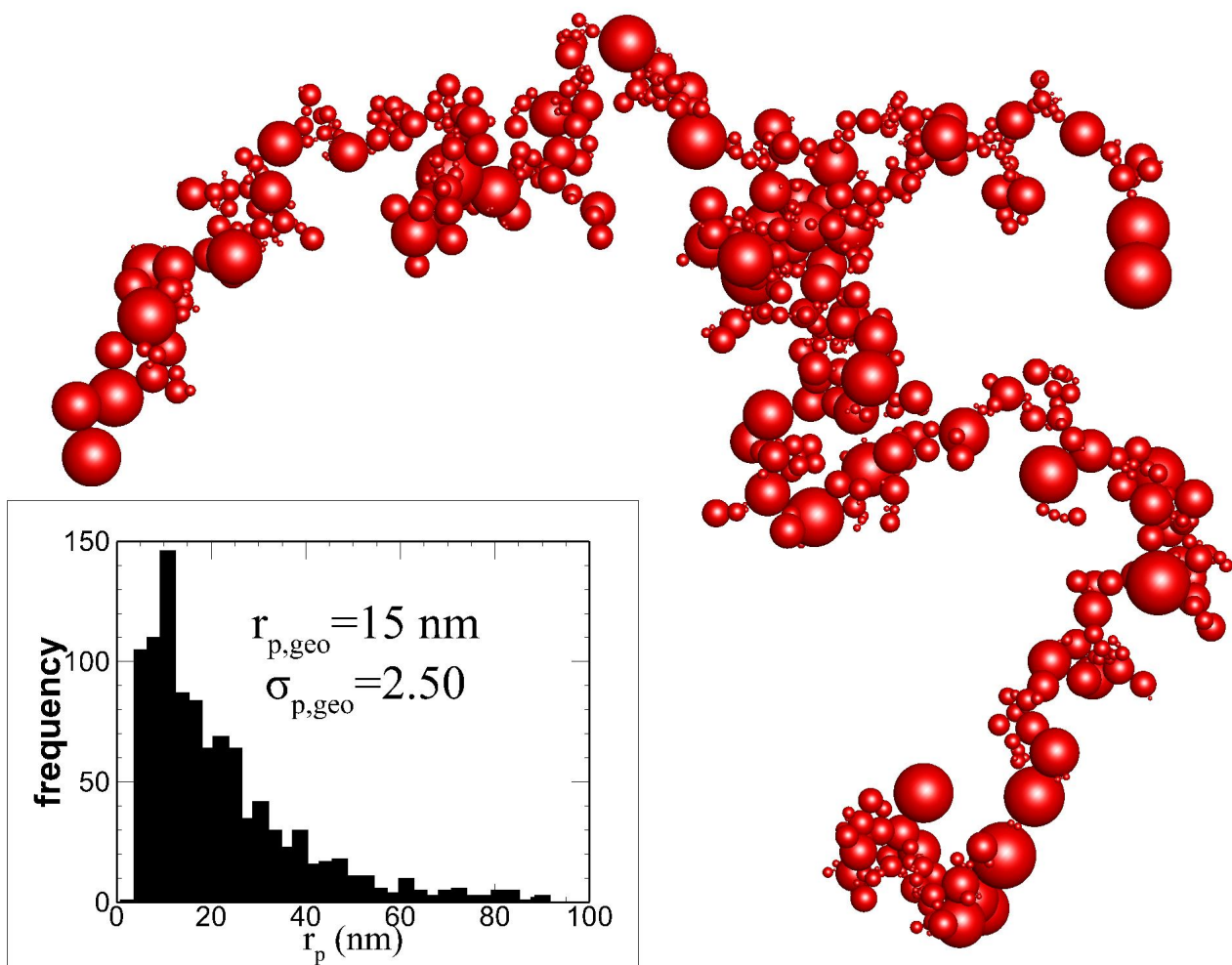


Figure 9: Example of aggregate consisting of $N=1024$ monomers and $\sigma_{p,geo} = 2.50$.

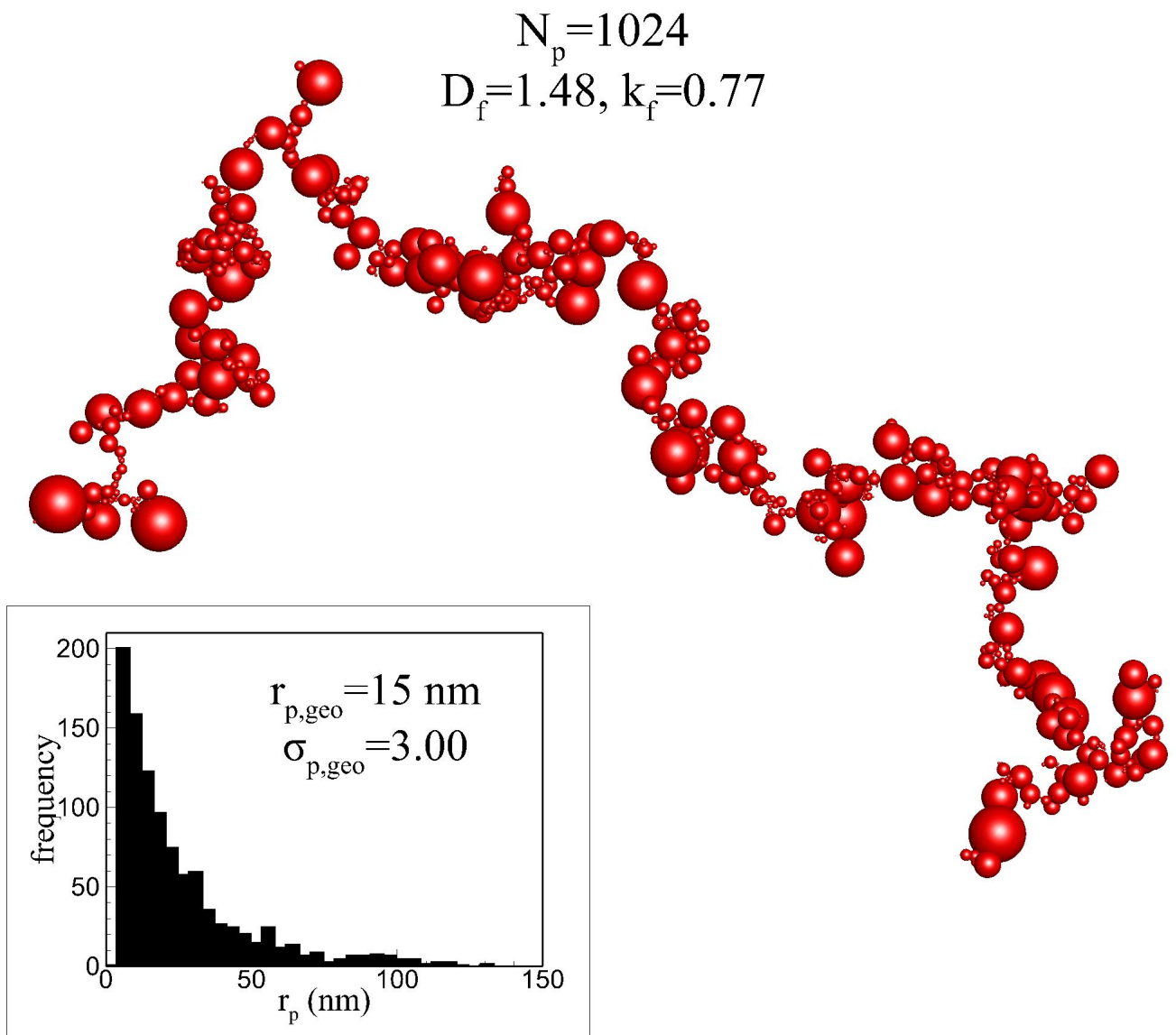


Figure 10: Example of aggregate consisting of $N=1024$ monomers and $\sigma_{p,geo} = 3.00$.